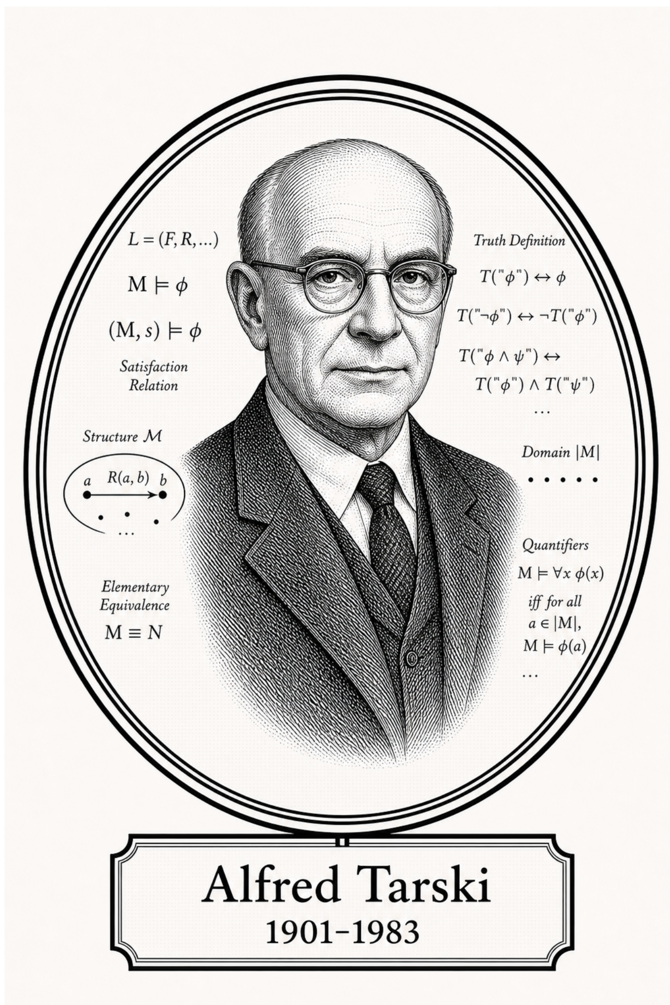
From Cantor to Ito

Advanced Logic

Model Theory



Alfred Tarski

1901-1983

Self-Study Notes

William Sollers

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For every reader learning to make mathematics their own.

Contents

Chapter 1

Model Theory

1.1 Languages and Structures

L-Structures Toolkit — Quick Reference

Concept	Symbol / Notation
First-order language	$\mathcal{L}$
L-structure	$\mathcal{M} = (M, \mathcal{L}^{\mathcal{M}})$
Domain / universe	$\lvert\mathcal{M}\rvert$ or M
Interpretation of f	$f^{\mathcal{M}} : M^n \to M$
Interpretation of R	$R^{\mathcal{M}} \subseteq M^n$
Interpretation of c	$c^{\mathcal{M}} \in M$
Variable assignment	$s : \mathrm{Var} \to M$
Term evaluation	$s(t) \in M$
Satisfaction	$\mathcal{M} \models \varphi[s]$

First-Order Languages

Definition (First-Order Language $\mathcal{L}$)

Definition 1.1.1 (First-Order Language $\mathcal{L}$)**.** A **first-order language** $\mathcal{L}$ consists of three (possibly empty) sets of non-logical symbols:

- a set $\mathcal{R}$ of **relation symbols** (also called *predicate symbols*), each assigned a fixed arity $n \geq 1$;
- a set $\mathcal{F}$ of **function symbols**, each assigned a fixed **arity** $n \geq 0$ (0-ary function symbols are *constant symbols*);
- a set $\mathcal{C}$ of **constant symbols** (equivalently, 0-ary function symbols — listed separately for emphasis).

Together with the logical symbols common to all first-order languages (variables, connectives $\neg, \wedge, \vee, \to, \leftrightarrow$, quantifiers $\forall, \exists$, equality $=$, parentheses), the triple $(\mathcal{F}, \mathcal{R}, \mathcal{C})$ determines $\mathcal{L}$.

Remark (Dependencies).

- Arity
- Relation Symbol
- Function Symbol
- Constant Symbol
- Variable
- Logical Connective

Remark (Fully quantified form). $\mathcal{L} = (\mathcal{F}, \mathcal{R}, \mathcal{C})$ where $\mathrm{ar}(f) \in \mathbb{N}$ for each $f \in \mathcal{F}$ and $\mathrm{ar}(R) \in \mathbb{N}_{\geq 1}$ for each $R \in \mathcal{R}$.

Remark (Intuition). A language is a *vocabulary*: it names the operations, relations, and distinguished elements that can appear in sentences, without yet saying what those names mean. The meaning is supplied by a structure.

Remark (Source note). Some sources (e.g. Marker §1.1, Hodges) absorb constant symbols into 0-ary function symbols. Others (e.g. Chang–Keisler) list them separately. The content is equivalent; only the bookkeeping differs.

Remark (Consequence). The set of $\mathcal{L}$-terms and the set of $\mathcal{L}$-formulas are both defined inductively from $\mathcal{L}$; see term evaluation below.

Example (Standard examples of first-order languages).

1. **Language of ordered fields** $\mathcal{L}_{\mathrm{of}}$: function symbols $\{+, \cdot, -\}$ of arities $2, 2, 1$; relation symbol $\{<\}$ of arity 2; constant symbols $\{0, 1\}$. The real field $(\mathbb{R}, +, \cdot, -, <, 0, 1)$ is an $\mathcal{L}_{\mathrm{of}}$-structure.

2. **Language of graphs** $\mathcal{L}_{\text{gr}}$: no function symbols, no constants; one binary relation symbol $\{E\}$ (for "edge").

3. **Language of groups** $\mathcal{L}_{\text{gp}}$: function symbols $\{\cdot, {}^{-1}\}$ of arities $2, 1$; constant symbol $\{e\}$.

4. **Pure equality language** $\mathcal{L}_{=}$: no non-logical symbols at all. Structures are arbitrary non-empty sets.

L-Structures

Definition ($\mathcal{L}$-Structure)

Definition 1.1.2 ($\mathcal{L}$-Structure)**.** Let $\mathcal{L} = (\mathcal{F}, \mathcal{R}, \mathcal{C})$ be a first-order language. An **$\mathcal{L}$-structure** $\mathcal{M}$ is an interpretation of the symbols of $\mathcal{L}$ on a non-empty domain.

Remark (Structural data)*.* An **$\mathcal{L}$-structure** $\mathcal{M}$ consists of:

1. a non-empty set M called the **domain** (or **universe**) of $\mathcal{M}$, written $|\mathcal{M}|$;
2. for each n-ary relation symbol $R \in \mathcal{R}$, a relation $R^{\mathcal{M}} \subseteq M^n$;
3. for each n-ary function symbol $f \in \mathcal{F}$, a function $f^{\mathcal{M}} : M^n \to M$;
4. for each constant symbol $c \in \mathcal{C}$, a distinguished element $c^{\mathcal{M}} \in M$.

The superscript notation $f^{\mathcal{M}}, R^{\mathcal{M}}, c^{\mathcal{M}}$ denotes the **interpretation** of each symbol in $\mathcal{M}$.

Remark (Dependencies)*.*

- First-Order Language
- Arity
- Relation Symbol
- Function Symbol
- Constant Symbol
- Relation
- Function
- Cartesian Product
- Subset

Remark (Fully quantified form)*.* $\mathcal{M}$ is an $\mathcal{L}$-structure iff $M \neq \emptyset$, and $\forall f \in \mathcal{F}_{(n)},\ f^{\mathcal{M}} : M^n \to M$, and $\forall R \in \mathcal{R}_{(n)},\ R^{\mathcal{M}} \subseteq M^n$, and $\forall c \in \mathcal{C},\ c^{\mathcal{M}} \in M$.

Remark (Intuition). A structure is an $\mathcal{L}$-vocabulary *made concrete*: each symbol is assigned an actual mathematical object of the correct type. The domain provides the raw material; the interpretations give the symbols their meaning inside that domain.

Remark (Common error). The domain M must be *non-empty*. First-order logic with an empty domain is pathological: $\forall x\, \varphi$ would be vacuously true and $\exists x\, \varphi$ vacuously false for every φ, breaking the duality of the quantifiers.

Remark (Source note). Some sources write $\mathfrak{M}$ (Fraktur) for structures and reserve M for the domain; others use $\mathcal{M}$ throughout and write $|\mathcal{M}|$ for the domain. The latter convention is used here. Marker uses $\mathcal{M}$; Enderton uses $\mathfrak{A}$.

Example (L-structures for familiar mathematical objects). 1. $(\mathbb{R}, +^{\mathbb{R}}, \cdot^{\mathbb{R}}, -^{\mathbb{R}}, <^{\mathbb{R}}, 0^{\mathbb{R}}, 1^{\mathbb{R}})$ is an $\mathcal{L}_{\text{of}}$-structure with domain $M = \mathbb{R}$.

2. $(\mathbb{Q}, +^{\mathbb{Q}}, \cdot^{\mathbb{Q}}, -^{\mathbb{Q}}, <^{\mathbb{Q}}, 0^{\mathbb{Q}}, 1^{\mathbb{Q}})$ is another $\mathcal{L}_{\text{of}}$-structure with domain $M = \mathbb{Q}$.

3. The complete graph K_3 on $\{1, 2, 3\}$ is an $\mathcal{L}_{\text{gr}}$-structure with $E^{K_3} = \{(1, 2), (2, 1), (1, 3), (3, 1), (2, 3), (3,$

Two different structures can share the same language — the language only fixes the *type* of the interpretations, not their specific content.

Variable Assignments

Definition (Variable Assignment)

Definition 1.1.3 (Variable Assignment)**.** Let $\mathcal{M}$ be an $\mathcal{L}$-structure with domain M, and let $\mathrm{Var} = \{v_1, v_2, v_3, \ldots\}$ be a fixed countable set of first-order variables.
A **variable assignment** (or **environment**) for $\mathcal{M}$ is a function $s : \mathrm{Var} \to M$.
For a variable assignment s, a variable v_i, and an element $a \in M$, write $s[v_i \mapsto a]$ for the assignment that agrees with s on all variables except v_i, where it returns a:

$$s[v_i \mapsto a](v_j) = \begin{cases} a & \text{if } j = i, \\ s(v_j) & \text{if } j \neq i. \end{cases}$$

Remark (Dependencies).

- $\mathcal{L}$-Structure
- Variable
- Function
- Countable

Remark (Fully quantified form). $s : \mathrm{Var} \to M$, and $s[v_i \mapsto a](v_j) = a$ if $j = i$, else $s(v_j)$.

Remark (Intuition). A variable assignment gives temporary values to free variables before evaluating a formula. The $s[v_i \mapsto a]$ notation is the standard tool for handling quantifier rules: to check $\exists v_i\, \varphi$, test whether $\mathcal{M} \models \varphi[s[v_i \mapsto a]]$ for some $a \in M$.

Term Evaluation

Definition (Term Evaluation)

Definition 1.1.4 (Term Evaluation)**.** Let $\mathcal{M}$ be an $\mathcal{L}$-structure and $s : \mathrm{Var} \to M$ a variable assignment. The **evaluation** $s(t) \in M$ of an $\mathcal{L}$-term t under s is defined inductively:

1. If $t = v_i$ is a variable, then $s(t) = s(v_i)$.
2. If $t = c$ is a constant symbol, then $s(t) = c^{\mathcal{M}}$.
3. If $t = f(t_1, \ldots, t_n)$ for an n-ary function symbol f, then $s(t) = f^{\mathcal{M}}(s(t_1), \ldots, s(t_n))$.

Remark (Dependencies).

- $\mathcal{L}$-Structure
- Variable Assignment
- Term
- Variable
- Arity
- Function Symbol
- Constant Symbol

Remark (Fully quantified form). $s : \mathrm{Var} \to M$, and $s(t) \in M$ for every $\mathcal{L}$-term t. Formally: $s(\cdot)$ is the unique function on terms satisfying clauses (1)–(3).

Remark (Intuition). Term evaluation is the "computation" step: given concrete values for all variables, every term reduces to an element of M. The inductive definition mirrors the inductive definition of terms themselves.

Satisfaction [Stub]

Definition (Satisfaction, $\mathcal{M} \models \varphi[s]$) — Stub

Let $\mathcal{M}$ be an $\mathcal{L}$-structure, s a variable assignment, and φ an $\mathcal{L}$-formula. The **satisfaction relation** $\mathcal{M} \models \varphi[s]$ (read: "$\mathcal{M}$ satisfies φ under s") is defined inductively on the structure of φ:

[To be filled in — clauses for atomic formulas, negation, conjunction, disjunction, implication, existential quantifier, universal quantifier.]

Remark (Fully quantified form). *[Stub — to be completed.]*

Remark (Intuition). Satisfaction is the semantic notion of truth: $\mathcal{M} \models \varphi[s]$ means φ is true in $\mathcal{M}$ when each free variable v_i takes the value $s(v_i)$. The definition proceeds by induction on formula complexity, grounding out on atomic formulas evaluated via term evaluation.

Elementary Equivalence and Substructures [Stub]

Proofs

Capstone

Bibliography